

FDE Take-Home Assessment: Motion Graphics Pipelines

(~4 Hours)

Overview

Client: Yosuki Musical Instrument Corporation

Brief: Yosuki Musical Instrument Corporation, a leading Japanese manufacturer of guitars, pianos, and saxophones, regularly utilizes After Effects templates in their marketing workflows. Generally, their existing pipelines allow their creative team to easily utilize existing product photography, product renders, and taglines to create localized and variant-driven ads. They produce 970x250px billboard banner ads, 16:9 YouTube ads, and 1:1 sponsored Instagram reels for their catalog of musical instruments.

Product Lines: The client has chosen three product lines to implement upgrades within their pipeline: saxophones, pianos, and guitars. Yosuki sells a single model of saxophone, three different models of piano, and three different models of guitars which come in two different colors.

Task: Implement a pipeline which utilizes the brand's existing assets and an input brief to implement motion graphics templates for banner ads. Final rendered assets should be 5-10s long. Example assets should be provided in a zip file by your Talent Coordinator *or* can be entirely replaced by assets of your own choosing if you prefer.

Product Lines	Models	Variants	Subject Variation Requirements	Aspect Ratios	Minimum Outputs
Yosuki Saxophone	1	1 Each	One Model of Saxophone, No Variations	Billboard 16:9 1:1	3
Yosuki Pianos	3	1 Each	Three Different Models of Piano, No Variations	Billboard 16:9	6
Yosuki Guitars	3	2 Each	Three Different Models of Guitar, Two Colors	Billboard 16:9 1:1	18

Requirements

Input Layer

Utilize the provided campaign brief to extract a structured representation (JSON or CSV) of the required data for successful pipeline execution. Use structured input (JSON or CSV) to represent creative brief constraints & expectations (eg. brand, product, target audience, creative tone, tagline, etc.)

Implement a structured AI generation layer that:

- Produces structured JSON output from a brief or prompt
- Enforces schema validation or structured parsing
- Generates a product tagline, CTA copy, and creative instruction
- Demonstrates prompt strategy and variation logic
- Handles potential text overflow constraints

Creative Generation Layer

Yosuki has provided 3D renders, as well as existing product photography to guide a generative model(s) on final product placement in a scene for the ad. These can either be a generated image or video for the background of the final ad.

Implement a creative AI generation layer that uses:

- creative instruction from the Input Layer's output to form a prompt for direction
- provided product images **and/or** renders as a reference for the final generated background asset.
- a consistent output destination for the next layer to use
- **optionally**: the 3D object placed in a virtual stage, assembled according to the creative instruction, and used as reference for the final generated background asset

Automation & Orchestration Layer

Build an automation & orchestration layer that:

- Clearly separates input > AI generation > templated transformation > render > delivery
- Populates the After Effects template with each variant's content
- Supports batch processing
- Eliminates manual copy & paste steps
- Delivers final rendered assets to a consistent output destination (Dropbox, Google Drive, etc.)

Deliverables & Tooling

The final solution must be provided with:

- A public GitHub repository if:
 - implemented as a software solution
 - using any custom services/nodes for n8n, ComfyUI, etc.
 - contains any custom plugins or ExtendScript you developed for After Effects
- Narrated walkthrough and screen capture of the solution working end-to-end.

Non-functional submissions will not be evaluated further.

Required: Please record a quick demo of the exercise working to help the team set up/run the app locally for review. You will be asked to submit the assignment and demo for review **before** we schedule any follow-up interviews.

If the assignment passes, you will be asked to share the information in a 20-30 minute walk-through conversation with 2 team members.